

Hamza Abbas

Lecturer, Faculty of Mechanical Engineering | GIK Institute, Topi, KPK, Pakistan
hamza.abbas0133@gmail.com | +92 348 0555875 | [linkedin.com/in/hamza-abbas1](https://www.linkedin.com/in/hamza-abbas1)

Profile

Mechanical engineering lecturer and researcher at GIK Institute, specialising in thermofluids, porous-media transport phenomena, and digital microfluidics. MS thesis developed a modified Lucas–Washburn governing equation for heterogeneous porous assemblies (validated to $\pm 8\%$ error). Industrial project experience includes a full Aspen Plus ORC+VARS techno-economic model (8.7 MW, 17.9% combined efficiency) for DG Khan Cement Company. Hands-on digital microfluidics competency spans EWOD contact-angle tuning, CO₂-laser micromachining, photolithographic electrode patterning, and thermal lamination. Two journal articles under review at *Langmuir* (ACS) and *Journal of Fluid Mechanics* (Cambridge University Press).

Work Experience

University Lecturer – *GIK Institute, Faculty of Mechanical Engineering* Sep 2024 – Present
Topi, Pakistan

- Courses: Refrigeration & Air-Conditioning / HVAC, Engineering Dynamics, CAD/CAM Lab.
- Lab instruction: Thermofluids-I; Lab Incharge, Fluid Mechanics Laboratory.
- LMS Coordinator; Outcome-Based Education (OBE) mapping aligned to Washington Accord.
- Supervised accreditation visit (February 2026): lab readiness, alumni/employer documentation.
- Co-ordinated 6th International Workshop on Functional Reverse Engineering of Machine Tools (WRE 2024), December 2024.

Graduate Research Assistant – *Biofuels & Microfluidics Lab, GIK Institute* Feb 2022 – May 2024
Topi, Pakistan

- Led experimental and computational work on paper-based microfluidics, lab-on-a-chip devices, passive micropumps, and MEMS.
- Derived and validated a modified Lucas–Washburn equation for fluid transport in heterogeneous porous assemblies (permeability-ratio and interfacial-contact factors; $\pm 8\%$ error).
- Mentored undergraduates on COMSOL Multiphysics heat-and-mass-transfer modelling.
- Lab instructor: Thermo-Fluid Lab 1. Course assistance: Heat Transfer, Measurement & Instrumentation, Micro & Nano Fabrication.

Project Research Assistant (HEC Project TTST-74) – *GIK Institute / FME* Feb 2022 – Feb 2024
Topi, Pakistan

- Higher Education Commission Pakistan project: *High-Resolution Flexible Time-Temperature Indicator (FRTTI) for Food Quality Control*.
- Developed Business Model Canvas for the “Flow Time” startup in collaboration with Dr Ali Turab Jafry; early engagement with milk-trader stakeholders.

Mechanical Engineering Intern – *DG Khan Cement Company Limited* Aug 2020 – Sep 2020
Chakwal, Pakistan

- Studied plant schematic diagrams; participated in maintenance activities for cement and power-plant machinery.

- Maintenance of limestone crusher, raw mill, and kiln; quality-control department operations.

Education

M.S. in Mechanical Engineering (Thermofluid Specialisation) – GIK Institute Feb 2022 – Jun 2024
Topi, Pakistan

- GPA: 3.67 / 4.00. Merit-Based Graduate Scholarship & Assistantship (GA-1).
- Thesis: *Dynamics of Fluid Flow Between Two Porous Membranes*.
- Key contribution: modified Lucas–Washburn equation with dimensionless permeability-ratio (z) and interfacial-contact factor (λ) for stratified porous assemblies.

B.S. in Mechanical Engineering – GIK Institute
Topi, Pakistan

Aug 2017 – Jul 2021

- Senior Design Project: *Design and Analysis of a Waste Heat Recovery Unit (ORC integrated with VARS) for DG Khan Cement Company* – Aspen Plus modelling, Peng-Robinson EOS; combined efficiency 17.9%, net output 8.7 MW.

Research Projects

Industrial Water Heating System – Pakistan Tobacco Company (PTC) May 2025 – Present
Design of a large-scale industrial water heating system integrated with solar PV; full technical and cost report in progress.

ORC Feasibility Study – Low-Temperature Heat Source Feb 2025 – Mar 2025
Techno-economic feasibility of an Organic Rankine Cycle for a low-grade industrial heat source.

ORC+VARS Waste Heat Recovery – DG Khan Cement Sep 2020 – Jul 2021
Replaced hazardous Ammonia-based Kalina cycle with an integrated ORC+VARS system. Aspen Plus (Peng-Robinson EOS) simulation; *n*-Hexane working fluid selected after comparative analysis with R245fa and ammonia; combined efficiency 17.9%, net turbine output 8.7 MW.

COMSOL Packed-Bed Latent-Heat Storage Simulation Feb 2020 – Jun 2020
Course project simulating a packed-bed latent heat storage tank using COMSOL Multiphysics.

Integrated Solar-ORC-LiBr Absorption Chiller (Aspen Plus) Sep 2020 – Jan 2021
Modelling of an integrated system based on a solar heat source, organic Rankine cycle, and water/LiBr absorption chiller.

Publications

Journal Articles – Under Review

[2025] Abbas, H.; Jafry, A.T.; Ali, M.; Ajab, H.; Cheema, T.A. “Impact of paper thickness, contact area, and fluid volume on flow dynamics in paper-based microfluidics.” **Langmuir (ACS)**, Under review.

[2025] Ali, M.; Jafry, A.T.; Abbas, H.; Lee, M.; Abbasi, M.S.; Shah, S.F.; Ajab, H. “Revealing the Mechanics of Highly Resistive Fluid Flow in a Leaf Architecture using Paper Microfluidics.” **Journal of Fluid Mechanics (Cambridge)**, Under review.

- [2024] Ullah, H.; ...; Abbas, H.; ... "Droplet Actuation Enhancement through Voltage Control and Hydrophobic Coating Selection." **MATEC Web of Conferences**, Vol. 398, p. 01020, EDP Sciences.
- [2023] Abbas, H.; Ali, M.; Ullah, H.; Jafry, A.T. "Actuation and Flow Control in Paper-Based Microfluidics by Varying Thickness of Storage Reservoir." **MATEC Web of Conferences**, Vol. 381, p. 01010, EDP Sciences.
- [2023] Abbas, H.; Ali, M.; Naeem, N.; Ullah, H.; Jafry, A.T. "Effect of Storage Reservoir on Fluid Velocity in Lateral Flow Paper Device." **Engineering Proceedings (MDPI)**, 45(1), 5.
- [2023] Ali, M.; Jafry, A.T.; Abbas, H. "Flow Control in Paper-Based Microfluidic Channel by using Hydrophobic Tape Barrier on the Paper Surface." **MATEC Web of Conferences**, Vol. 381, p. 01009, EDP Sciences.
- [2023] Ali, M.; Ullah, H.; Ali, M.; Naeem, N.; Abbas, H.; ... "Dynamic Contact Angle Variation with Applied Voltage and Droplet Volume in Digital Microfluidics." **Engineering Proceedings (MDPI)**, 45(1), 31.
- [2023] Ullah, H.; Shoaib, ...; Abbas, H.; Jafry, A.T. "Design and Fabrication of Digital Microfluidics Device for Lab-on-a-Chip Applications." **Engineering Proceedings (MDPI)**, 45(1), 7.
- [2022] Ali, M.; Hussain, G.; Abbas, H.; Jafry, A.T. "Flow Control in Paper-Based Microfluidics Using Variable Porosity Channel." **Engineering Proceedings (MDPI)**, 23(1), 33.

Technical Skills

Process Simulation	Aspen Plus (steady-state, Peng-Robinson & SRK EOS, cycle optimisation, heat & mass balance)
Multiphysics / CFD	COMSOL Multiphysics (heat transfer, fluid dynamics, MEMS, porous media)
Fabrication	CO ₂ laser micromachining, photolithographic electrode patterning, thermal lamination
Data & Analysis	Origin-Pro (curve fitting), ImageJ (optical flow tracking, contact-angle measurement)
Programming	Python (basic); MATLAB
Office / LMS	Microsoft Office; Moodle LMS; OBE documentation (Washington Accord)
Languages	Urdu / Punjabi (native); English (C1 – CEFR)

Professional Memberships & Awards

- Pakistan Engineering Council – Registered Mechanical Engineer (Lifetime member), Sep 2021.
- 5th place, DICE AFS-2022 Innovation Competition, MNS University of Agriculture, Multan, Oct 2022.
- GIK Institute Merit-Based Graduate Scholarship & Assistantship (GA-1), Feb 2022.

Conferences & Professional Development

- Conference Organiser, 2nd International Conference on Modern Technologies in Mechanical & Material Engineering, GIK Institute, Apr 2024.
- 3-Day Workshop on Nanomaterial Characterisation Techniques (XRD, SEM, AFM, FTIR, Raman, Hall Effect), USPCAS-E NUST, Islamabad, Sep 2022.

- Continuing Professional Development: Academic Writing; Multi-Body Dynamics (Adams); Laser-Based Fabrication for Microfluidics – GIK Institute.
- OBE System Development Programme, GIK Institute, 2022.

References

Dr Ali Turab Jafry

Associate Professor, FME
GIK Institute, Topi
ali.turab@giki.edu.pk

Dr Taqi Ahmad Cheema

Dean / Associate Professor, FME
GIK Institute, Topi
tacheema@giki.edu.pk